

Cryptology Assignment - 4

classmate

Date

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- ① cipher text : HLDTOIHJDXWQCMAN
key = $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$ use $A=0, B=1, \dots$ find plain text.

→ find determinant :
 $|K| = (3)(7) - (2)(5)$
 $= 21 - 10 = \underline{11}$

→ $11^{-1} \text{ mod } 26 = 19$

→ find Inverse matrix :

$$K^{-1} = 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \text{ mod } 26$$

$$K^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

HL :

$$C = \begin{bmatrix} 7 \\ 11 \end{bmatrix}$$

$$P = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 11 \end{bmatrix} = \begin{bmatrix} 21 + 154 \\ 63 + 55 \end{bmatrix} = \begin{bmatrix} 175 \\ 118 \end{bmatrix}$$

$$175 \text{ mod } 26 = 19 = T$$

$$118 \text{ mod } 26 = 14 = O$$

DT :

$$C = \begin{bmatrix} 3 \\ 19 \end{bmatrix}$$

$$P = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 19 \end{bmatrix} = \begin{bmatrix} 9 + 266 \\ 27 + 95 \end{bmatrix} = \begin{bmatrix} 275 \\ 122 \end{bmatrix}$$

$$275 \text{ mod } 26 = 15 = P$$

$$122 \text{ mod } 26 = 18 = R$$

Q1

$$C = \begin{bmatrix} 16 \\ 8 \end{bmatrix}$$

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$$P = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 16 \\ 8 \end{bmatrix} = \begin{bmatrix} 160 \\ 180 \end{bmatrix}$$

$$160 \bmod 26 = 4 = E$$

$$180 \bmod 26 = 2 = C$$

HT

$$C = \begin{bmatrix} 7 \\ 9 \end{bmatrix}$$

$$P = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 9 \end{bmatrix} = \begin{bmatrix} 147 \\ 108 \end{bmatrix}$$

$$147 \bmod 26 = 12 = R$$

$$108 \bmod 26 = 4 = E$$

DX

$$C = \begin{bmatrix} 3 \\ 23 \end{bmatrix}$$

$$P = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 23 \end{bmatrix} = \begin{bmatrix} 331 \\ 142 \end{bmatrix}$$

$$331 \bmod 26 = 19 = T$$

$$142 \bmod 26 = 12 = M$$

WQ

$$C = \begin{bmatrix} 22 \\ 16 \end{bmatrix}$$

$$P = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 22 \\ 16 \end{bmatrix} = \begin{bmatrix} 290 \\ 278 \end{bmatrix}$$

$$290 \bmod 26 = 4 = E$$

$$278 \bmod 26 = 18 = S$$

CM

$$C = \begin{bmatrix} 2 \\ 12 \end{bmatrix}$$

$$P = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 2 \\ 12 \end{bmatrix} = \begin{bmatrix} 174 \\ 78 \end{bmatrix}$$

$$174 \bmod 26 = 18 = R$$

$$78 \bmod 26 = 0 = A$$

AG: $C = \begin{bmatrix} 0 \\ 6 \end{bmatrix}$

$$P = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 0 \\ 6 \end{bmatrix} = \begin{bmatrix} 84 \\ 30 \end{bmatrix}$$

$$84 \bmod 26 = 6 = G$$

$$30 \bmod 26 = 4 = E$$

→ TOP SECRET MESSAGE

(2) consider Hill cipher key matrix

$$\begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix}$$

cipher text: RXADY

→ find plain text for above cipher text.

Determinant:

$$|K| = 24 \times 14 - 19 \times 5$$

$$= 336 - 95 = 241$$

$$\rightarrow 241 \bmod 26 = 7$$

$$7^{-1} = 15$$

Inverse matrix

$$K^{-1} = 15 \begin{bmatrix} 14 & -19 \\ -5 & 24 \end{bmatrix} \bmod 26$$

$$K^{-1} = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix}$$

Rx

$$\begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 17 \\ 23 \end{bmatrix} = \begin{bmatrix} 57 \\ 557 \end{bmatrix}$$

$$57 \bmod 26 = 5 = F$$

$$557 \bmod 26 = 17 = R$$

AD

$$C = \begin{bmatrix} 0 \\ 14 \end{bmatrix}$$

$$P = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 0 \\ 14 \end{bmatrix} = \begin{bmatrix} 14 \\ 308 \end{bmatrix}$$

$$14 \bmod 26 = 14 = O$$

$$308 \bmod 26 = 22 = W$$

DY

$$C = \begin{bmatrix} 3 \\ 24 \end{bmatrix}$$

$$P = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 3 \\ 24 \end{bmatrix} = \begin{bmatrix} 30 \\ 537 \end{bmatrix}$$

$$30 \bmod 26 = 4 = E$$

$$537 \bmod 26 = 17 = R$$

→ FLOWER

④ Transposition Encryption

Encrypt "The arrow is pointing north direction" using
Simple, with key = 4

T H E A
R R O W
I S P O
I N T I
W N N O
R T H O
I R E E
T O O N

→ cipher text: TRIINRITHRSPNONTREIOPNTNHE
OAWOIODCN

→ keyword

using key = ATLAS.

① ⑤ ③ ① ④
A T L A S
T H E A R
R O W I S
P O I N T
I N G N O
R T H D I
R E C T I
O N X X X

ciphertext: TRPIRROA INNDTX LEWIGHCX
STOIIX HOONTEN

→ double : key 1 = Break
key 2 = Log

B R E A K	L O G
2 5 3 1 4	2 3 1
② ⑤ ③ ① ④	② ③ ①
B R E A K	L O G
T H E A R	A I N
R O W I S	N D T
P O I N T	X T R
I N G N O	P I R
R T H D I	R O E
R E C T I	W I G
O N X X X	H C X
	A R T

→ AINNDTX TAPIRROEWI O I I
GHCX STOIX HOONT
EN
ONT
ENX

→ NTRREGXTIDTX
ANXPR WHROXOE
IDTIDICSIHNN

→ Decryption with key 10 (simple).

TIDIE MRNME REDGO SANDW RBLTP EEDDO
SSANO AHTUD BIENP GODAE PEIXD ELNXX.

→ key = 10, character = 60

$$\rightarrow \frac{60}{10} = 6$$

T	R	O	O	P	S	H	E	A	D
I	N	G	W	E	S	T	N	E	E
D	M	O	R	E	S	U	P	P	L
I	E	S	S	E	N	D	G	E	N
E	R	A	L	D	U	B	O	I	S
M	E	N	T	O	A	I	D	X	X

TROOPS HEADING WEST NEED MORE
SUPPLIES SEND GENERAL DUBOIS TO AID.

→ keyword
SECRET

ciphertext: TLEDT NHPIR BOIT MAXXA
XERXT ANONI SNFOT X

$$\rightarrow \frac{36}{6} = 6$$

1	T	L	E	D	T	N
2	A	P	I	R	S	O
3	I	I	T	M	A	X
4	S	A	X	E	R	X
5	T	A	N	O	N	I
6	S	N	F	O	T	X

5	2	1	4	3	R
S	E	C	R	E	T

	L	E	C	R	E	T
T	H	I	L	I	L	
A	P	L	A	I	N	
N	T	E	X	T	F	
O	R	O	T	O	R	
N	L	N	E	N	M	
I	O	X	X	X	T	

THIS IS A PLAIN.

→ keyword - PLAIN
cipher - VEY EAX ARA ATX HGD

PLAIN → A I L N P.

$$\frac{15}{5} = 3$$

P	L	A	I	N	
H	A	V	E	A	Y E Y E A
G	R	E	A	T	X
D	A	Y	X	X	

→ HAVE A GREAT DAY

→ Permutation - Find inverse permutation of

① 4 3 1 5 2
 1 2 3 4 5

→ $P^{-1} = 3, 5, 2, 1, 4$

② 7 4 1 3 5 2 6
 1 2 3 4 5 6 7

$P^{-1} = 3, 6, 4, 2, 5, 7, 1$

③ 4 1 3 2
 1 2 3 4

$P^{-1} = 2, 4, 3, 1$

④ 3 5 1 4 2
 1 2 3 4 5

$P^{-1} = 3, 5, 1, 4, 2$

Classical Ciphers: Confusion vs Diffusion

Cipher	Does it change the character/symbol value?	Does it change the character position?	Confusion / Diffusion	Justification
Shift Cipher	Yes	No	Confusion	Each plaintext character is replaced by another character by shifting its value. The position of characters remains unchanged.
Multiplicative Cipher	Yes	No	Confusion	Each character value is multiplied by a key modulo 26. The character positions are not changed.
Affine Cipher	Yes	No	Confusion	Each character value is transformed using $C = (aP + b) \bmod 26$. The positions of characters remain unchanged.
Transposition Cipher	No	Yes	Diffusion	The character values remain the same, but their positions are rearranged according to a permutation/key.